

Final answers (record here — graded on the answer; show your work):

1. _____ 2. _____ 3. _____ 4. _____ 5. _____ 6. _____

Symmetry: Even, Odd, or Neither

Example (Even, odd, or neither):

$f(x) = x^3$ gives $f(-x) = (-x)^3 = -x^3 = -f(x)$, so it is **odd** (origin symmetry).

Directions: Find $f(-x)$, then classify as even, odd, or neither.

1. $f(x) = x^2 + 1$

4. $f(x) = |x|$

2. $f(x) = x^3 - x$

5. $f(x) = x^5$

3. $f(x) = x^2 + x$

Challenge Problems

6. $f(x) = x^4 - 3x^2$ — test for even/odd and explain.

Answer Key and Procedure

Procedure

1. Substitute $-x$ for x .
2. Simplify carefully (watch the signs on odd powers).
3. If it equals $f(x)$ it is even; if it equals $-f(x)$ it is odd; otherwise neither.

Answers

1. Even
2. Odd
3. Neither
4. Even
5. Odd
6. Even (all terms have even powers)